Characterization of Cascaded LTI Instances Using Mutual Information: a Computer Study

Aman Chawla

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Abstract

In this paper, the authors carry out some investigations on LTI systems with random noise. Putting them into long cascades, they study the input-output relationship through the lens of mutual information. The authors find patterned formations of mutual information with respect to system parameters.

LTI systems are the bread and butter of linear systems theory. We follow the lead of [1] and characterize an LTI system by developing a cascade of the same system type (see Figure 1), with randomness included at one (numerator) parameter of the system (out of four possible parameters). The chain is "stimulated" at the left end point. The input is taken at the output of the first LTI system in the chain and the output is taken from the last LTI system. This mimics the myelinated axon studied in the reference above.

One expects variable performance due to noise and no systematic differences between instances. However, one finds patterns in the mutual information (as characterized in Figures 2 through 6) outcomes. These will be investigated theoretically in future work. This study has much potential for novel characterization of nonlinear systems as well. Further, there is no particular reason to confine attention to cascades and one can look at other configurations such as feedback and general LTI networks.

References

[1] Aman Chawla. Using Mutual Information to Characterize Neural PDEs. July 2024.

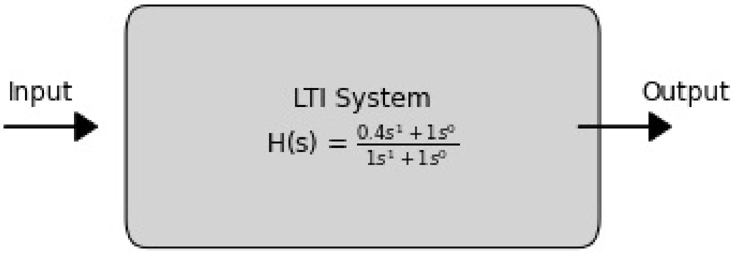

Figure 1: Instance of the simple LTI system forming cascades, studied numerically in this paper.

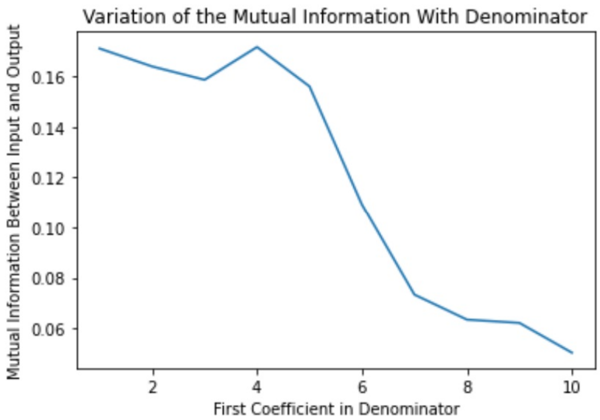

Figure 2: Mutual Information and First Denominator Coefficient. Each data point is a summary of 100 runs.

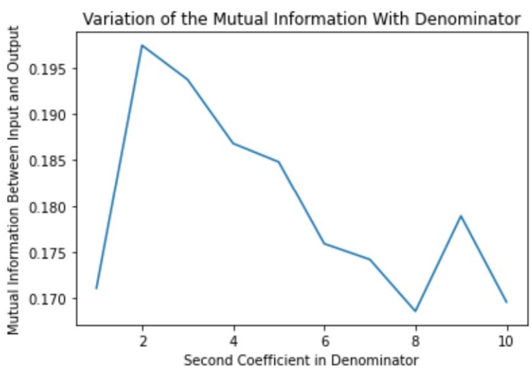

Figure 3: Mutual Information and Second Denominator Coefficient. Each data point is a summary of 100 runs.

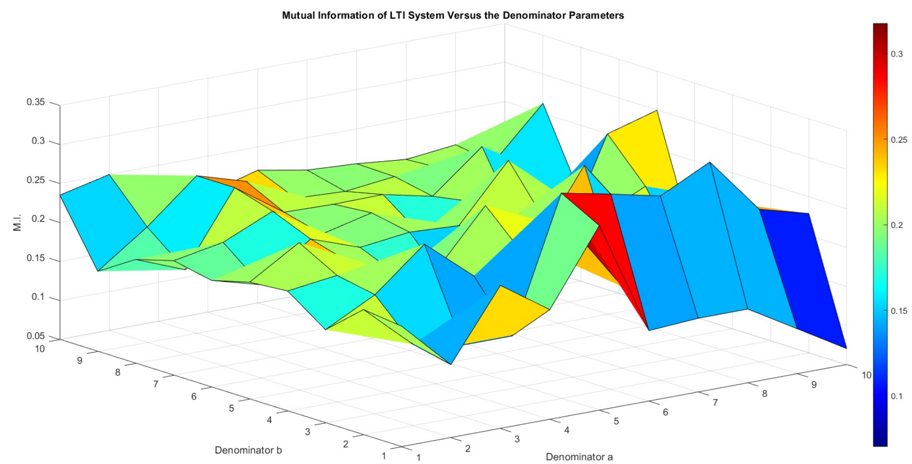

Figure 4: Variation of m.i. with both denominator parameters. Each data point is a summary of 2 runs.

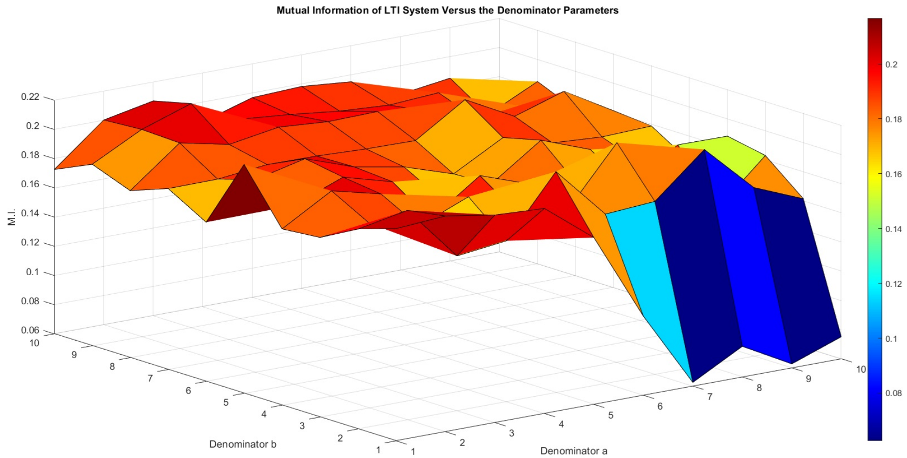

Figure 5: Variation of m.i. with both denominator parameters. Each data point is a summary of 20 runs.

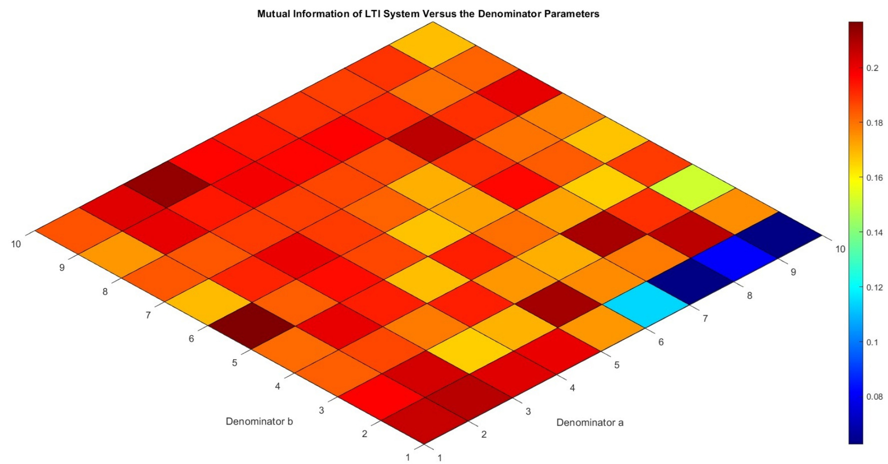

Figure 6: Variation of m.i. with both denominator parameters: Second View. Each data point is a summary of 20 runs. An “upper triangular” pattern is clearly visible, where the color map is at a higher temperature or higher mutual information, on average, as compared to the opposite “lower triangle.”